

Question No 5

Problem Analysis Chart

Data	Processing	Output	Decision
Total vehicles N	1) Initialize capacities and set all entries to 0	→ per vehicle	• Main loop: runs N times
Vehicle Type (C/B/V)	2) loop N times to process each car	• Assigned zones & remaining space or rejection reason	• Validation loop repeats until valid inputs are entered
Category (F/S/G)	3) Validate in inner loop to check valid input	2) Final Summary • c_tot, c_acc, c_aj • c_car, c_bikes, c_van	
Permit P (Y/N)	4) Check permit. if invalid, ask if emergency vehicle, if not emergency reject	• Occupied space left in A, B, C • Zone with highest Occupancy	→ Capacity rule • Zone A max = 80 • Zone B max = 40 • Zone C max = 15
Emergency flag m (Y/N, if permit N)	5) Check eligibility for emergency flag 6) Check zone capacity	• Capacity full alert	• Space usage • Van in zone C takes 2 spaces • All other valid entries take 1 space
	a) Faculty (F) → zone A b) Student (S) → zone B (if van redirected to zone C) c) Guest (G) → zone C (van needs ≥ 2 spaces)	8) if no space, reject and state reason	
	7) Update zone occupancy & vehicle counters	a) Display final statistics and facility status	

2. Input Processing Output.

Input:

- $n_{in} = \text{vehicles}$
- $v_type (C, B, V)$
- $user_cat (F, S, G)$
- $has_permit (Y, N)$
- $is_emergency (Y, N)$

all valid

closed

Output:

rule

max = 30

min = 40

max = 15

usage

zone

spaces

valid

1 space

ad

lie

- Vehicle Status
- (Assigned zone or rejection reason)
- Remaining zone capacity
- Final summary report

Processing

1) Initialize capacities and counters

2) For vehicle = 1 to n_{in} vehicles

a) Validate inputs (v_type , $user_cat$, has_permit).

b) If $is_emergency$

- $is_emergency$

- If NOT $emergency$ Then reject

c) Determine required spaces

- Van = 2 spaces, Car/Bike = 1 space

d) Determine preferred zone

based on user category & rules

e) If capacity available then

- Update zone capacity

- Increment accepted counters

Else

- Reject (No available space)

3) Compute highest occupancy zone

4) Compute campus full status

3- Algorithm:

1. Initialize Constants and Variables

- Set Zone Max capacities $CAP_A = 20, CAP_B = 40, CAP_C = 15$
- Set zone Occupancies: $occ_A = 0, occ_B = 0, occ_C = 0$
- Set counters: $count_{cars} = 0, count_{bikes} = 0, count_{vars} = 0$
 $Total_accepted = 0, Total_rejected = 0$

2. Input Total Vehicles: Prompt security officer to enter total vehicle to process (N)
Validate $N > 0$

3) Process Each Vehicle (loop for $i = 1$ to N)

a) Validate Vehicle Type (v_type)

- Prompt for vehicle (C for Car, B for Bike, V for Van)
- loop until v_type is valid.

b) Validate User Category ($user_cat$)

- Prompt for category (F for faculty, S for student, O for Visitor / Guest)
- loop until $user_cat$ is valid.

c) Validate Permit (has_permit)

- Prompt for permit (Y for Yes, N for No)
- loop until has_permit is valid

3) Evaluate Eligibility and Imaging status

- $Sid_is_image = N$
- If has $perit == N$, pass for imaging (Y/N) with validation
- If $(has_perit == N)$ AND $image == N$:
 - Reject vehicle with reason "Invalid Perit or Non Imaging"
 - Increment total rejected. Continue to next vehicle iteration

4) Determine Required Spaces and Target Zones

- Determine required spaces: $req_spaces = 2$ if $V_type == 'V'$ else 1.

Zone Selection Rules:

- Faculty ($user_cat == 'F'$): Target Zone A
- Student ($user_cat == 'S'$):
 - if $V_type == 'V'$, Target Zone C (Student was redirected to Visitor zone)
 - Else, Target Zone B.
- Visitor ($user_cat == 'G'$): Target Zone C.

5) Zone Capacity Check & space Allocation

- If Target zone is A?
 - If $(CAP_A - occ_A) \geq req_spaces$, park in Zone A ($occ_A += req_spaces$)
 - Else, reject vehicles with reason "No available space in Zone A"

- If Target zone is B:
 - If $(CAP_B - occ_B) \geq req_spaces$
park in zone B ($occ_B += req_spaces$)
 - Else, reject vehicle with reason "No available space in zone B"
- If Target zone is C:
 - if $(CAP_C - occ_C) \geq req_spaces$
park in zone C ($occ_C += req_spaces$)
 - Else, reject vehicle with reason "No available space in zone C"

6) Update Counters.

- if assigned successfully
 - Increment Total accepted
- if $V_type == 'C'$
 - increment count cars
- if $V_type == 'B'$
 - increment count Bikes
- if $V_type == 'V'$
 - increment count vans
- Display assigned zone & remaining capacity for that zone
- if rejected
 - Increment Total - rejected
 - Display rejection reason

7) Display Summary Report.

- Compute ready capacity?

$$\text{rem}_A = \text{CAP}_A - \text{occ}_A,$$

$$\text{rem}_B = \text{CAP}_B - \text{occ}_B,$$

$$\text{rem}_C = \text{CAP}_C - \text{occ}_C.$$

- Determine zone with highest occupancy among zone A, B & C

- Determine if entire campus is full
($\text{rem}_A == 0 \text{ AND } \text{rem}_B == 0 \text{ AND } \text{rem}_C == 0$)

- Print msg